

Green, Gauss, Stokes, and the Bridge to Series

Lecture 12 for Multivariable Calculus

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Lecture Plan

Lecture 12 closes the vector-calculus part of the course and opens the next large topic. The guiding theme is that a complicated boundary integral can often be replaced by an easier interior integral.

1. Review scalar line integrals, work integrals, surface area, and flux.
2. Use Green's theorem for circulation and flux in the plane.
3. Use the divergence theorem to turn a closed surface flux into a triple integral.
4. Use Stokes' theorem to turn a space-curve circulation into a surface integral.
5. Compare the three big theorems as one boundary-interior principle.
6. Begin infinite sequences and series: partial sums, geometric series, p -series, and first tests.

Central Message

The vector-calculus theorems have the same shape:

$$\boxed{\text{integral on the boundary} = \text{integral of a derivative inside.}}$$

The derivative may be $Q_x - P_y$, $\operatorname{div} F$, or $\operatorname{curl} F$. The boundary may be a closed curve or a closed surface. The practical skill is choosing the side that is easier to compute.

1 Warm-Up: Which Integral Is It?

Before using a theorem, identify the object being integrated.

$$\int_C f \, ds, \quad \int_C F \cdot dr, \quad \iiint_S f \, dS, \quad \iint_S F \cdot n \, dS.$$

These four expressions answer different questions.

Integral	Geometric object	Meaning
$\int_C f \, ds$	curve C	mass of a wire, weighted length
$\int_C F \cdot dr$	oriented curve C	work, circulation
$\iiint_S f \, dS$	surface S	mass of a sheet, surface area
$\iint_S F \cdot n \, dS$	oriented surface S	flux through a surface

Parametrization Formulae

For a curve $r(t)$, $a \leq t \leq b$,

$$dr = r'(t) dt, \quad ds = \|r'(t)\| dt.$$

Thus

$$\int_C F \cdot dr = \int_a^b F(r(t)) \cdot r'(t) dt, \quad \int_C f ds = \int_a^b f(r(t)) \|r'(t)\| dt.$$

For a surface $X(u, v)$,

$$dS = \|X_u \times X_v\| du dv, \quad n dS = \pm(X_u \times X_v) du dv.$$

Orientation check. Scalar integrals with ds or dS do not depend on orientation. Vector integrals with dr or $n dS$ do depend on orientation.

2 Green's Theorem

Let D be a plane region and let ∂D be its positively oriented boundary. Positive orientation means that the region stays on the left as one walks around the boundary.

Theorem 2.1 (Green's Theorem: Circulation Form). *If P, Q have continuous first partial derivatives on a region containing D , then*

$$\oint_{\partial D} P dx + Q dy = \iint_D (Q_x - P_y) dA.$$

Meaning

The quantity $Q_x - P_y$ measures infinitesimal rotation in the plane. Green's theorem says that the total circulation around the boundary equals the accumulated microscopic rotation inside.

2.1 Example: Circulation Around a Circle

Compute

$$\oint_C -y dx + x dy,$$

where C is the circle $x^2 + y^2 = a^2$ oriented counterclockwise.

Here $P = -y$ and $Q = x$, so

$$Q_x - P_y = 1 - (-1) = 2.$$

By Green's theorem,

$$\oint_C -y \, dx + x \, dy = \iint_{x^2+y^2 \leq a^2} 2 \, dA = 2\pi a^2.$$

Direct check. Parametrize C by

$$r(t) = (a \cos t, a \sin t), \quad 0 \leq t \leq 2\pi.$$

Then

$$dx = -a \sin t \, dt, \quad dy = a \cos t \, dt,$$

and

$$-y \, dx + x \, dy = -(a \sin t)(-a \sin t) \, dt + (a \cos t)(a \cos t) \, dt = a^2 \, dt.$$

Therefore the integral is

$$\int_0^{2\pi} a^2 \, dt = 2\pi a^2.$$

2.2 Green's Theorem: Flux Form

For a plane vector field $F = (P, Q)$, the outward flux across ∂D is

$$\oint_{\partial D} F \cdot n \, ds.$$

If ∂D is counterclockwise, then the outward normal element is

$$n \, ds = (dy, -dx).$$

Thus

$$F \cdot n \, ds = P \, dy - Q \, dx.$$

Applying Green's theorem to $(-Q, P)$ gives

$$\boxed{\oint_{\partial D} F \cdot n \, ds = \iint_D (P_x + Q_y) \, dA = \iint_D \operatorname{div} F \, dA.}$$

Example. Let $F = (x, y)$ and let D be the disk $x^2 + y^2 \leq a^2$. Since

$$\operatorname{div} F = 1 + 1 = 2,$$

the outward flux is

$$\oint_{\partial D} F \cdot n \, ds = \iint_D 2 \, dA = 2\pi a^2.$$

Common Mistake

Green's theorem needs a positively oriented boundary. If the boundary is clockwise, the circulation form changes sign. The flux form uses the outward normal, so check the normal direction separately.

3 Divergence Theorem

Green's flux formula is the two-dimensional version of the divergence theorem.

Theorem 3.1 (Divergence Theorem). *Let $E \subset \mathbb{R}^3$ be a solid with closed boundary ∂E , oriented by the outward unit normal n . If F is continuously differentiable, then*

$$\iint_{\partial E} F \cdot n \, dS = \iiint_E \operatorname{div} F \, dV.$$

Physical Meaning

The divergence $\operatorname{div} F$ measures local source strength. If $\operatorname{div} F > 0$, the field is locally flowing outward. If $\operatorname{div} F < 0$, the field is locally flowing inward. The total outward flux through the boundary equals the total source strength inside.

3.1 Example: Flux Through a Sphere

Let

$$F(x, y, z) = (x, y, z)$$

and let S be the sphere $x^2 + y^2 + z^2 = a^2$ oriented outward. Compute

$$\iint_S F \cdot n \, dS.$$

Since

$$\operatorname{div} F = 1 + 1 + 1 = 3,$$

the divergence theorem gives

$$\iint_S F \cdot n \, dS = \iiint_{x^2+y^2+z^2 \leq a^2} 3 \, dV = 3 \cdot \frac{4}{3}\pi a^3 = \boxed{4\pi a^3}.$$

Direct check. On the sphere, the outward unit normal is

$$n = \frac{(x, y, z)}{a}.$$

Thus

$$F \cdot n = \frac{x^2 + y^2 + z^2}{a} = a.$$

Therefore

$$\iint_S F \cdot n \, dS = a \cdot 4\pi a^2 = 4\pi a^3.$$

3.2 Example: Avoiding a Difficult Surface Integral

Let

$$F = (x^3, y^3, z^3)$$

and let S be the boundary of the cube

$$0 \leq x, y, z \leq 1.$$

The direct flux integral has six faces. The divergence theorem gives one triple integral:

$$\operatorname{div} F = 3x^2 + 3y^2 + 3z^2.$$

Thus

$$\iint_S F \cdot n \, dS = \int_0^1 \int_0^1 \int_0^1 (3x^2 + 3y^2 + 3z^2) \, dz \, dy \, dx.$$

By symmetry,

$$= 3 \cdot \int_0^1 \int_0^1 \int_0^1 3x^2 \, dz \, dy \, dx = 9 \int_0^1 x^2 \, dx = 3.$$

When to Use the Divergence Theorem

Use it when the surface is closed and the divergence is simpler than the flux integral. It is especially useful for boxes, balls, cylinders, and surfaces made of several pieces.

4 Stokes' Theorem

Stokes' theorem is the three-dimensional version of Green's circulation theorem.

Theorem 4.1 (Stokes' Theorem). *Let S be an oriented smooth surface with boundary curve ∂S . The orientation of ∂S is determined by the right-hand rule from the chosen normal n . If F is continuously differentiable, then*

$$\oint_{\partial S} F \cdot dr = \iint_S (\operatorname{curl} F) \cdot n \, dS.$$

Right-Hand Rule

Point the thumb of the right hand in the direction of the surface normal. The curled fingers give the positive boundary orientation.

4.1 Example: Replace a Space Curve by a Disk

Let

$$F = (-y, x, z)$$

and let C be the circle

$$x^2 + y^2 = a^2, \quad z = 0,$$

oriented counterclockwise as viewed from above. Compute

$$\oint_C F \cdot dr.$$

Choose S to be the disk $x^2 + y^2 \leq a^2$ in the plane $z = 0$, with normal $n = (0, 0, 1)$. Then

$$\text{curl } F = \begin{vmatrix} i & j & k \\ \partial_x & \partial_y & \partial_z \\ -y & x & z \end{vmatrix} = (0, 0, 2).$$

Therefore

$$\oint_C F \cdot dr = \iint_S (0, 0, 2) \cdot (0, 0, 1) \, dS = 2\pi a^2.$$

4.2 Same Boundary, Different Surface

Stokes' theorem lets us choose any convenient surface with the same boundary, as long as the orientation is consistent. If two surfaces S_1 and S_2 have the same boundary curve C , then

$$\iint_{S_1} (\text{curl } F) \cdot n \, dS = \iint_{S_2} (\text{curl } F) \cdot n \, dS$$

provided the two induced boundary orientations agree.

Common Mistake

Do not use the divergence theorem on an open surface. Do not use Stokes' theorem without checking the boundary orientation. These are orientation theorems first and computation tools second.

5 One Picture for the Big Theorems

The vector-calculus theorems are not separate tricks. They are different faces of one principle.

Theorem	Boundary integral	Interior derivative	Dimension
FTC	$f(b) - f(a)$	$f'(x)$	interval
Green circulation	$\oint_{\partial D} P \, dx + Q \, dy$	$Q_x - P_y$	plane region
Green flux	$\oint_{\partial D} F \cdot n \, ds$	$\operatorname{div} F$	plane region
Stokes	$\oint_{\partial S} F \cdot dr$	$\operatorname{curl} F$	surface in space
Divergence	$\iint_{\partial E} F \cdot n \, dS$	$\operatorname{div} F$	solid in space

Generalized Stokes Slogan

$$\int_{\partial\Omega} \omega = \int_{\Omega} d\omega.$$

The boundary operator ∂ and the derivative operator d are paired. This is why the phrase “boundary of boundary is zero” appears throughout vector calculus.

6 Problem-Solving Checklist

Choosing a Theorem

1. Is the integral over a closed plane curve? Try Green’s theorem.
2. Is the integral a flux through a closed surface? Try the divergence theorem.
3. Is the integral a circulation around a space curve? Try Stokes’ theorem.
4. Is the field conservative? Use a potential function instead.
5. Is the theorem side actually easier? If not, parametrize directly.

Orientation checklist.

- Green circulation: counterclockwise boundary is positive.
- Green flux: use outward normal.
- Divergence theorem: use outward normal on the closed surface.
- Stokes’ theorem: normal and boundary orientation must match the right-hand rule.

7 Bridge to Infinite Series

After integration, the course turns to another kind of accumulation: adding infinitely many terms.

New Question

For a sequence of numbers $a_1, a_2, a_3, \dots$, what does it mean to add all of them?

$$a_1 + a_2 + a_3 + \dots$$

The answer is not to “add forever.” The answer is to study the sequence of partial sums.

7.1 Sequences

A sequence is a function whose input is a positive integer:

$$n \mapsto a_n.$$

We say that a_n converges to L if

$$\lim_{n \rightarrow \infty} a_n = L.$$

For example,

$$\frac{1}{n} \rightarrow 0, \quad \left(1 + \frac{1}{n}\right)^n \rightarrow e, \quad (-1)^n \text{ does not converge.}$$

7.2 Series and Partial Sums

An infinite series is written as

$$\sum_{n=1}^{\infty} a_n.$$

Its N th partial sum is

$$s_N = \sum_{n=1}^N a_n.$$

The series converges to S if the sequence of partial sums converges:

$$\lim_{N \rightarrow \infty} s_N = S.$$

In that case,

$$\sum_{n=1}^{\infty} a_n = S.$$

Necessary but Not Sufficient

If $\sum a_n$ converges, then $a_n \rightarrow 0$. But $a_n \rightarrow 0$ does not guarantee convergence. The harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

diverges even though $\frac{1}{n} \rightarrow 0$.

7.3 Geometric Series

The geometric series is

$$\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + \cdots.$$

The partial sum is

$$s_N = a \frac{1 - r^{N+1}}{1 - r}, \quad r \neq 1.$$

If $|r| < 1$, then $r^{N+1} \rightarrow 0$, so

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1 - r}, \quad |r| < 1.$$

If $|r| \geq 1$, the geometric series diverges except for the trivial case $a = 0$.

Example.

$$1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \cdots = \frac{1}{1 - \frac{1}{2}} = 2.$$

7.4 p-Series

The p -series is

$$\sum_{n=1}^{\infty} \frac{1}{n^p}.$$

It converges if and only if

$$p > 1.$$

Thus

$$\sum_{n=1}^{\infty} \frac{1}{n^2} \text{ converges,} \quad \sum_{n=1}^{\infty} \frac{1}{n} \text{ diverges.}$$

7.5 First Tests

The first tests are simple but powerful.

Basic Series Tests

1. **Term test for divergence.** If $a_n \not\rightarrow 0$, then $\sum a_n$ diverges.
2. **Geometric test.** $\sum ar^n$ converges exactly when $|r| < 1$.
3. **p -series test.** $\sum 1/n^p$ converges exactly when $p > 1$.
4. **Comparison idea.** If $0 \leq a_n \leq b_n$ and $\sum b_n$ converges, then $\sum a_n$ converges. If $0 \leq b_n \leq a_n$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Summary

1. Green's theorem converts a plane boundary integral into a double integral.
2. The divergence theorem converts closed-surface flux into a triple integral.
3. Stokes' theorem converts boundary circulation into surface curl flux.
4. Orientation is part of the data, not decoration.
5. Infinite series are defined through partial sums.
6. The geometric series and p -series are the first models for convergence.